DSAA 4040 Course Project Proposal

E1 —Cloud-Native Online Bookstore on Kubernetes

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April 20, 2026

1. Track & Choice

Engineering-Oriented ·E1 —Cloud-Native Online Bookstore on Kubernetes. I want a project that ships something clickable, not just numbers in a table. E1 covers the full cloud-native loop —containerize, deploy, expose, scale, observe —which is the part of cloud-native I most want hands-on time with. Level-2 also feels like the right fit for a solo group: enough surface area to build a real system end-to-end without drowning in infra setup.

Team. Solo (1 person). I'll own frontend, backend, K8s manifests, observability, load testing, and the report.

2. Scope

A minimum-but-honest online bookstore —browse / search books, cart, place order —on single-node Kubernetes (Minikube). Vue 3 SPA, FastAPI backend, PostgreSQL for state, Redis in front of read-heavy endpoints. On top I wire **HPA + Prometheus + Grafana** so scaling decisions are measurable rather than just "it scaled". Explicit non-goals: real auth, real payments, multi-tenant admin.

3. Architecture (Frontend / Backend / Execution Layer)

Three layers. Frontend handles user traffic and static delivery; backend owns business logic and state; execution layer is the K8s platform —scheduling, scaling, config, storage, observability.

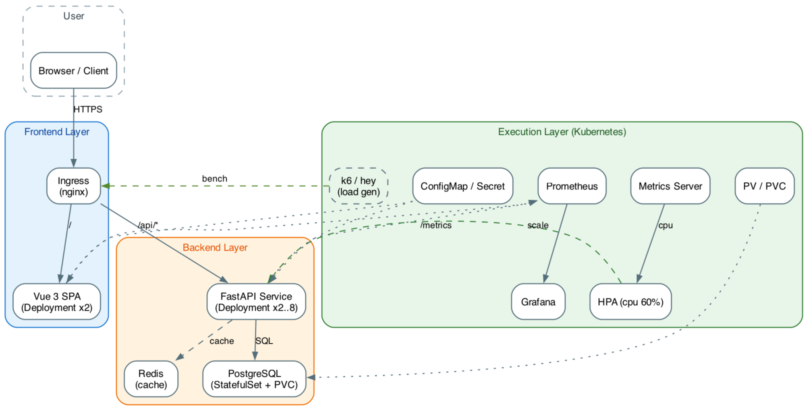

Figure 1: System architecture of the bookstore, organized as Frontend / Backend / Execution Layer on Kubernetes.

4. Preview of the Frontend

To sanity-check the UI direction before coding, I had Claude sketch a quick HTML mockup of the catalog page. It's not the final stack —just a concept shot so I know what I'm aiming for: a card grid for books, a top health strip showing pod count and p95 latency, and a sidebar with categories. Wiring the health badge to live cluster metrics during the demo is the part that ties the UI back to K8s state.

5. Task Levels

Basic + Standard + 3 Advanced. Basic: frontend/backend/DB containerized and deployed with Deployment + Service; browse/cart/order working. Standard: ConfigMap + Secret, Ingress, architecture diagram, one-command deploy. Advanced (3 of 4): (a) HPA with before/after under burst traffic; (b) Prometheus + Grafana dashboard; (c) k6 / hey performance test with latency and throughput numbers. Skipping "DB schema improvement"as a standalone advanced item —happens organically, won't pad the report.

6. Timeline (SOP)

Written as an SOP because what matters for solo execution is "what do I do next", not "where is the bar on week 3". Each step has an Enter / Do / Exit; if Exit isn't met, I don't advance.

- **SOP-01 ·Local skeleton ·Apr 20–24.** *Enter:* proposal approved. *Do:* scaffold Vue 3 SPA and FastAPI

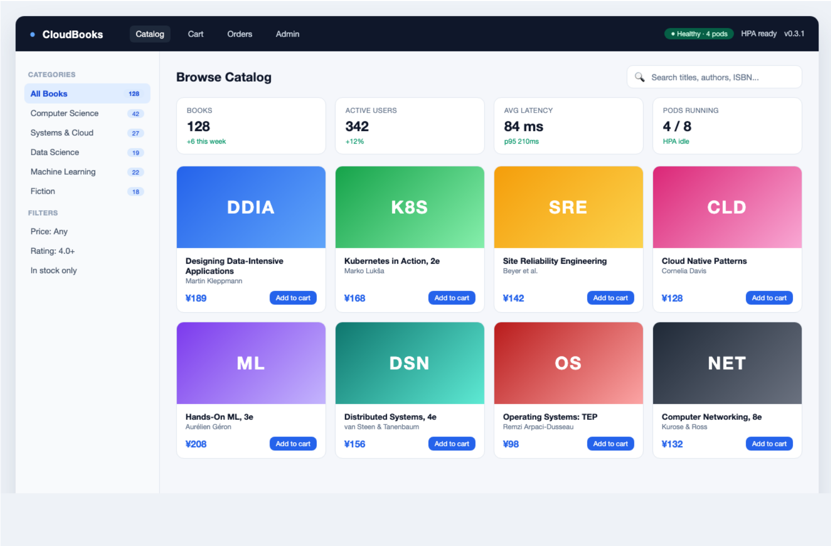

Figure 2: A quick catalog-page concept written by Claude for visual direction. Final UI will be built in Vue 3.

(`/health`, `/books`, `/cart`, `/orders`), Postgres via docker-compose, seed ~50 books. *Exit:* `docker compose up` serves a working browse -> cart -> order flow at `localhost:8080`.

- **SOP-02 ·Containerize ·Apr 25–26.** *Enter:* SOP-01 done. *Do:* multi-stage Dockerfiles, tag v0.1.0, load into Minikube. *Exit:* both images < 250 MB and runnable standalone with env vars.
- **SOP-03 ·First K8s deploy ·Apr 27–30.** *Enter:* SOP-02 done. *Do:* Namespace `bookstore`; Deployment + Service (frontend, backend); StatefulSet + PVC (Postgres); ConfigMap + Secret; Ingress. *Exit:* `kubectl apply -k .` on a clean Minikube yields a reachable site at `http://bookstore.local`.
- **SOP-04 ·Observability & HPA ·May 1–5.** *Enter:* SOP-03 done. *Do:* Metrics Server; HPA on API (min=2, max=8, CPU=60%); kube-prometheus-stack via Helm; expose FastAPI `/metrics`; Grafana dashboard (QPS, p95, pod count, CPU). *Exit:* sustained k6 load scales API pods 2 -> ≥4 visibly on Grafana in real time.
- **SOP-05 ·Mid-term demo ·May 6–7.** *Enter:* SOP-04 done. *Do:* 3–4 min video (clean apply -> browse -> trigger HPA -> Grafana) + status page. *Exit:* Milestone 2 submitted.
- **SOP-06 ·Load testing ·May 8–14.** *Enter:* M2 done. *Do:* three k6 scenarios —steady 50 RPS / burst 0->300 RPS in 30 s / sustained 150 RPS × 10 min; capture Grafana snapshots + CSV. *Exit:* one baseline-vs-HPA results table and three annotated screenshots, all reproducible from `scripts/bench/`.
- **SOP-07 ·Report & polish ·May 15–20.** *Enter:* SOP-06 done. *Do:* write report (Problem / Design / Implementation / Deployment / Evaluation / Limitations); polish README (one-command setup); record 5-min final demo. *Exit:* clone -> README -> running cluster on a fresh machine in under 15 min.
- **SOP-08 ·Submit ·May 21.** Tag v1.0.0, bundle source + report + video, submit.

7. Risks & Mitigations

Minikube quirks on macOS -> pin versions early, keep Docker Desktop K8s as fallback. Prometheus disk footprint -> minimal Helm values, disable Alertmanager. Solo timeline slip -> if SOP-06 slips I drop one k6 scenario rather than shorten SOP-07 polish. Scope creep -> non-goals above are firm.

8. Deliverables

Source code + Kustomize manifests + `docker-compose.yml`; one-command bootstrap (`make up`); PDF technical report; 5-minute final demo video; README with setup, reproduction, and cleanup instructions.